

Mid-Point Review Report

Automated File Analysis Tool for Windows Digital Forensics

By

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

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1. Introduction:

Digital forensic investigations increasingly encounter sophisticated anti-forensic techniques employed by cybercriminals to obstruct evidence collection and analysis. The proliferation of file extension spoofing, steganography, and timestamp manipulation presents significant challenges to forensic examiners working with large datasets. Traditional manual examination methods prove time-intensive and susceptible to human error, while existing forensic tools frequently lack integrated detection capabilities for multiple anti-forensic methods operating simultaneously.

Problem Statement:

The problem being addressed concerns the inadequate automation in detecting anti-forensic techniques within Windows file systems. Digital forensic investigators are affected by this limitation, particularly those working in law enforcement agencies, corporate security teams, and forensic laboratories. The problem occurs during the initial triage phase of investigations when examiners must rapidly assess large volumes of files to identify potential evidence.

The importance of solving this problem stems from several factors. Anti-forensics techniques target the availability, usefulness, and scientific validity of digital evidence, thereby complicating forensic analysis processes. Studies indicate that sophisticated obfuscation methods are increasingly prevalent in cybercrime investigations. Manual detection of these techniques across datasets containing thousands of files proves impractical within typical investigation timeframes. Furthermore, the absence of integrated tools forces investigators to employ multiple disparate applications, increasing analysis time and the potential for overlooked evidence.

The solution proposed involves developing an automated file analysis tool specifically for Windows operating systems. This tool will detect hidden files and steganography through signature analysis and entropy examination, providing investigators with automated detection capabilities and comprehensive reporting to accelerate digital forensic examinations while maintaining forensic integrity and chain-of-custody requirements.

Aims:

The project encompasses three primary aims:

1. To design an automated file analysis tool for Windows operating systems that detects anti-forensic techniques through multi-method analysis
2. To implement integrated detection capabilities combining file signature analysis, entropy examination, and metadata verification in a unified workflow
3. To develop comprehensive reporting and visualization features that communicate findings effectively to forensic investigators

Objectives:

Seven specific objectives have been established to achieve these aims:

1. To implement a file signature analysis module using magic number comparison for detecting extension-content mismatches

2. To develop a Shannon entropy calculator for identifying encrypted, compressed data and steganography indicators
3. To create an EXIF metadata parser capable of detecting timestamp anomalies and manipulation evidence
4. To build a cryptographic hash verification system supporting MD5 and SHA-256 algorithms for file integrity validation
5. To implement forensic evidence logging incorporating write-blocking mechanisms to maintain chain-of-custody
6. To design an interactive visualization dashboard displaying entropy distributions and timeline correlations
7. To perform usability testing with forensic practitioners measuring task completion time and detection accuracy

2. Background and Context:

The proliferation of digital devices in criminal activities has created an escalating demand for effective digital forensic analysis tools. Computer forensics is defined as the application of computer investigation and analysis techniques to gather evidence suitable for presentation in a court of law. The goal of computer forensics involves carrying out a structured investigation while documenting a chain of evidence to discover exactly what occurred on a computer system and who was responsible.

The emergence of anti-forensic techniques poses significant challenges to investigators. Anti-forensic methods include encryption, steganography, secure deletion, timestamp manipulation, and file signature manipulation. These techniques are employed by cybercriminals to conceal illicit activities, destroy evidence, and impede forensic examinations. The Windows NTFS file system, which dominates enterprise and personal computing environments, contains multiple locations where evidence may persist even after attempted destruction, including MFT, Log files, and locations.

The proposed tool addresses the critical need for automated detection mechanisms that can process large volumes of files efficiently while identifying multiple categories of anti-forensic manipulation. The target users include law enforcement agencies, corporate incident response teams, and forensic laboratories requiring rapid triage capabilities during investigations. The tool will operate within the Windows environment, examining files without modifying original evidence, thereby maintaining forensic integrity throughout the analysis process.

Summary of Relevant Published Work (Literature Review):

File Signature Analysis Techniques

File signature analysis represents a fundamental technique in digital forensics whereby file headers and extensions are compared against a database to identify hidden data. This method exposes discrepancies between file signatures and extensions, revealing attempts at file type obfuscation. The technique, also known as "magic number" analysis, examines the first several bytes of a file to determine its true format regardless of the file extension assigned.

The National Institute of Standards and Technology (NIST) method provides a standardized approach to analyze the authenticity of digital data during forensic investigations. Research by Mukaila et al. demonstrates that file signatures serve as forensic markers revealing attempts to disguise file types.

A common anti-forensic tactic where malicious actors rename executable files with innocuous extensions to evade detection. The study indicates that signature file analysis successfully identifies authentic file types in cases of information leakage, utilizing tools such as Access Data FTK Imager for evidence processing.

Modern forensic tools utilize signature databases containing thousands of known file formats. The research indicates that file signature search techniques are critical for identifying unknown or hidden files where extensions have been changed or removed. However, the literature also reveals limitations: signature analysis alone cannot detect sophisticated steganography or encryption within correctly identified file types.

Entropy-Based Detection Methods

Shannon entropy serves as a mathematical measure of randomness and information content within data. Research demonstrates that Shannon entropy can be used for detecting anomalies in images where steganography may be present. High entropy values typically indicate encrypted, compressed, or steganographically modified files, distinguishing them from unmodified content.

Meléndez et al. conducted experiments on steganography detection using entropy measures, establishing that redundant bits in image and audio files provide opportunities for information hiding. The research tested detection programs on 1,800 images using four different steganography algorithms, achieving 65% accuracy with false-positive rates less than 0.001%. However, Karthik and Kumar note that no universal technique currently exists that can detect steganography in any image with high accuracy and low computational requirements.

The study by Bat-Erdene et al. on entropy analysis for packer detection demonstrates that encrypted file header fragments exhibit unique Shannon entropy characteristics. This property was used to distinguish encrypted files from other high-entropy files such as archives. The research proposes enhancing entropy-based detection by combining three entropy algorithms and generating test cases based on threat tree models, thereby improving malicious file identification accuracy.

Critical evaluation of the literature reveals that entropy-based methods face challenges with false positives. Legitimate compressed files and high-quality JPEG images may exhibit entropy values similar to encrypted or steganographic content. The research suggests that multi-factor analysis combining entropy with other detection methods reduces false alarm rates.

Metadata and Timestamp Forensics

EXIF (Exchangeable Image File Format) metadata contains valuable forensic information including timestamps, device identifiers, GPS coordinates, and software details. Research by Oh et al. on forensic detection of timestamp manipulation demonstrates that investigators must examine multiple timestamp sources within Windows systems (Created, Modified, Accessed, Entry Modified) to identify inconsistencies suggesting manipulation.

The study by Hermon et al. on forensic recovery of file system metadata indicates that evidence may remain in file system locations even after deletion attempts. Research findings show that data wiping tools leave traces in NTFS locations including MFT and Logfiles, indicating that files or data may have been present on the disk previously. This discovery provides investigators with methods to detect anti-forensic activity even when primary evidence has been removed.

Markova et al. demonstrate that detection of relevant digital evidence in forensic timelines requires correlation analysis of metadata across multiple sources. The research indicates that timestamp manipulation represents a significant anti-forensic technique used to obscure the timeline of criminal activities. However, the presence of multiple timestamp records in different file system structures enables detection of manipulation attempts through inconsistency analysis.

Anti-Forensic Techniques and Countermeasures

Research by Park et al. on anti-forensic trace detection proposes a signature-based detection method for triage purposes that assists investigators in quickly making decisions about suspect digital devices before conducting full investigations. The study categorizes anti-forensic techniques into data hiding, artifact wiping, trail obfuscation, and attacks against forensic tools.

Al-Dhaqm et al. conducted a systematic literature review on countering anti-forensic tactics, identifying that encryption represents the most effective anti-forensic tool category since forensic analysis cannot be conducted on encrypted files without proper decryption keys. However, the research notes that the mere presence of encryption tools, encrypted containers, and patterns of encrypted files can serve as investigative leads.

The study by Markova et al. on the impact of anti-forensic techniques demonstrates that selected techniques such as file deletion, log deletion, alternate data streams (ADS) hiding, encryption, and fake evidence creation significantly impact forensic investigation integrity and algorithmic anomaly detection accuracy. The research proposes quantitative evaluation using precision, recall, and F1 score metrics to assess the measurable effect of anti-forensic techniques on investigation quality and reliability.

Research indicates that anti-forensic tools can be detected through various methods: encryption detection depends on identifying tool installation files and configuration information; steganography detection depends on statistical analysis, and trail obfuscation requires keeping systems updated to detect exploits. The literature emphasizes that understanding these impacts enables development of more robust detection mechanisms.

Forensic Datasets and Validation Requirements

The Computer Forensic Reference Data Sets (CFReDS) project provides documented sets of simulated digital evidence for tool validation, equipment checkout, investigator training, and proficiency testing. Research by Garfinkel et al. emphasizes that bringing science to digital forensics requires standardized forensic corpora. These datasets enable reproducible evaluation of forensic tools and techniques across multiple scenarios.

Grajeda et al. conducted a study on the availability of datasets for digital forensics, revealing gaps in publicly available corpora. The research suggests that while repositories like CFReDS, Digital Corpora, and Datasets for Cyber Forensics provide valuable resources, continued development of realistic datasets remains necessary to support evolving forensic tool development.

Summary of Literature Findings

The literature review reveals several key findings:

1. File signature analysis proves effective for detecting basic file type obfuscation but requires comprehensive signature databases and cannot detect sophisticated content-level manipulation
2. Entropy-based detection shows promise for identifying steganography and encryption but suffers from false positives, with no single universal solution achieving high accuracy across all scenarios
3. Metadata examination provides timeline reconstruction capabilities essential for detecting manipulation attempts, though analysis must consider multiple timestamp sources
4. Integrated approaches combining multiple detection methods outperform single-technique tools in accuracy and reliability
5. Standardized datasets are crucial for validation but often lack realistic usage patterns and complete ground truth documentation

Competitor Analysis / Related Work:

Several existing tools provide partial solutions to the problems addressed by this project:

Access Data Forensic Toolkit (FTK)

FTK represents a comprehensive commercial forensic suite offering file signature analysis, hash matching, and data visualization capabilities. The tool analyzes different file systems while revealing various timestamps. However, FTK's entropy analysis capabilities are limited, and the tool requires substantial licensing costs making it inaccessible to smaller organizations. The interface complexity also presents challenges for investigators requiring rapid triage functionality.

Autopsy/The Sleuth Kit

Autopsy provides open-source digital forensics capabilities including file signature verification and timeline analysis. Research indicates it represents a widely adopted tool for forensic examination. However, entropy-based detection requires manual configuration through third-party plugins, and integrated visualization of anti-forensic indicators remains limited. The tool excels at comprehensive examination but lacks specialized anti-forensic detection automation.

Oxygen Forensic Detective

Oxygen Forensic offers file signature search features with an extensive built-in database of file signatures for various file types. The tool generates detailed reports after identifying files with mismatched signatures and provides cross-platform analysis capabilities. However, the tool primarily targets mobile device forensics with limited focus on Windows file system anti-forensic detection. Entropy analysis and metadata correlation features remain underdeveloped compared to signature detection capabilities.

Tool	Signature Analysis	Entropy Detection	Metadata Analysis	Integrated Visualization	Cost
FTK	Comprehensive	Limited	Comprehensive	Basic Timeline	£3,500+
Autopsy	Comprehensive	Plugin required	Comprehensive	Good Timeline	Free
Oxygen	Excellent	Basic	Good	Limited	£1,500+

Proposed Tool	Comprehensive	Advanced	Comprehensive	Integrated	Free (Educational)
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The proposed tool differentiates itself through integrated anti-forensic detection combining signature analysis, entropy examination, and metadata correlation with specialized visualization designed specifically for identifying obfuscation attempts. The focus on Windows environments and automated triage functionality addresses gaps in current tool offerings.

3. Tools and Methods:

Rationale for Technical Tools Selected:

The technical implementation employs several tools and libraries selected based on specific requirements and capabilities:

Programming Language: Python

Python was selected as the primary development language due to its extensive ecosystem of forensic analysis libraries and rapid development capabilities. The language provides native support for file manipulation, cryptographic operations, and statistical calculations essential for the project. Python's interpreted nature enables iterative development and testing, accelerating the prototyping phase.

Alternative languages considered included C++ and Java. C++ was rejected despite superior performance characteristics due to longer development cycles and complexity in implementing cross-platform file system operations. Java was rejected due to JVM dependency requirements and complexity compared to Python's straightforward syntax for forensic operations.

Key Libraries and Frameworks:

- **python-magic (lib magic bindings):** This library provides access to the lib magic file type identification system, enabling detection of file types based on magic numbers rather than extensions. The library was chosen over alternatives like filetype due to its comprehensive signature database and active maintenance.
- **Pillow (PIL) and Exif Read:** These libraries enable EXIF metadata extraction from image files. Pillow provides image processing capabilities while Exif Read specializes in metadata parsing. The combination was selected to ensure robust metadata extraction across various image formats.
- **Hash lib (Python standard library):** The built-in hash lib module implements secure hash algorithms including MD5, SHA-1, and SHA-256. This library was chosen over third-party alternatives due to its native integration and cryptographic compliance with forensic standards.
- **NumPy and SciPy:** These scientific computing libraries support entropy calculations and statistical analysis. NumPy provides efficient array operations essential for processing large datasets, while SciPy offers statistical functions for analyzing detection results.
- **Matplotlib and Plotly:** These visualization libraries generate entropy distribution graphs and timeline visualizations. Matplotlib was selected for static report generation while Plotly provides interactive dashboard capabilities for real-time analysis.

Database System: SQLite

SQLite provides lightweight, serverless database capabilities suitable for storing analysis results, hash values, and evidence logs. The embedded database system was chosen over client-server alternatives like MySQL or PostgreSQL to eliminate external dependencies and simplify deployment. SQLite's transaction support ensures data integrity for forensic logging requirements.

User Interface Framework: Tkinter

Tkinter offers cross-platform GUI development with native Python integration. The framework was selected for its minimal external dependencies and straightforward deployment process. PyQt was considered as an alternative offering more advanced interface capabilities but rejected due to licensing complexities and increased application size.

Rationale for Research Methods Used:

NIST Digital Forensic Methodology

The project employs the NIST digital forensic methodology, which provides standardized procedures for analyzing digital data authenticity and obtaining valid evidence during identification processes. This methodology guides the development process through four phases:

1. **Collection:** Forensic imaging with write-blocking to preserve original evidence
2. **Examination:** Automated analysis of file signatures, entropy, and metadata
3. **Analysis:** Correlation of findings and timeline reconstruction
4. **Reporting:** Generation of court-admissible documentation

The NIST methodology was selected due to its widespread acceptance in forensic communities and alignment with legal requirements for evidence handling. Alternative methodologies considered included the ACPO (Association of Chief Police Officers) guidelines and the ISO/IEC 27037 standard. The NIST approach was chosen due to its comprehensive technical guidance and specific applicability to digital evidence analysis tools.

Project Management and Product Lifecycle:

Agile Development Process:

The project will follow the iterative development cycle with monthly milestones aligned with university assessment requirements. Development activities will be organized into weeks, each concluding with functional deliverables suitable for testing and evaluation.

Version Control and Documentation:

GitHub will serve as the primary version control repository with commit messages documenting development progress. The repository structure will include:

- /src (Source code for all modules)
- /tests (Unit tests and integration tests)
- /docs (Technical documentation and user guides)
- /data (Sample datasets and test cases)
- /results (Analysis outputs and validation results)

Commit messages will follow conventional commit format specifying feature additions, bug fixes, and documentation updates. This approach will enable clear tracking of development progress throughout the project lifecycle.

Project Management Tools:

GitHub Projects provides kanban board functionality tracking feature implementation, bug identification, and resolution progress. The board contains four columns: Backlog, In Progress, Testing, and Completed. Tasks are prioritized based on project dependencies and assessment milestone requirements.

Trello serves as an alternative task management platform with deadline tracking for major deliverables. The board includes cards for each objective with checklists breaking down implementation steps. Hence, GitHub suits the project most.

Risk Registered:

The following table identifies project risks, likelihood assessments, potential impacts, and mitigation strategies:

Risk Description	Likelihood	Impact	Mitigation Strategy
Dataset unavailability or access restrictions	Low	High	Multiple backup datasets identified (CFReDS and Corpora) also synthetic data generation capability developed
Detection algorithm inaccuracy exceeding acceptable thresholds	Medium	High	Early validation using known test cases, iterative refinement based on results; multiple detection methods to cross-validate findings
Performance issues with large datasets	Medium	Medium	Multi-threading implementation, progressive testing with increasing dataset sizes
Timeline overrun affecting milestone delivery	Medium	High	Buffer time incorporated in project plan; prioritization of core features over advanced features; regular supervisor communication maintaining alignment
Tool compatibility issues across Windows versions	Low	Medium	Testing on multiple Windows versions (10, 11); use of standard Python libraries avoiding version-specific dependencies
Library dependency conflicts or deprecation	Low	Low	Virtual environment isolation; documentation of library versions; monitoring for security updates and compatibility issues
False positive rate exceeding acceptable threshold	High	High	Multi-factor analysis combining signature, entropy, and metadata; configurable threshold parameters; extensive testing against diverse datasets

4. Requirements/Specifications:

Functional Requirements:

1: File Signature Analysis Capability

- The system shall identify file types using magic number analysis independent of file extensions
- The system shall compare detected file type with assigned file extension
- The system shall flag mismatches as potential obfuscation attempts with severity ratings
- The system shall support minimum 15 common file formats including JPEG, PNG, PDF, EXE, ZIP, DOCX, and compressed archives

2: Entropy Calculation and Analysis

- The system shall calculate Shannon entropy for each analyzed file
- The system shall flag files with entropy exceeding 7.5 as high-entropy content (potentially encrypted/compressed/steganographic)
- The system shall generate entropy distribution histograms for visual analysis
- The system shall analyze entropy patterns across file segments to detect embedded hidden data

3: Metadata Examination

- The system shall extract EXIF data from image files including timestamps, device information, and GPS coordinates
- The system shall identify timestamp inconsistencies such as Created timestamps exceeding Modified timestamps
- The system shall detect missing or suspicious metadata fields indicating potential manipulation
- The system shall analyze NTFS file system timestamps including Created, Modified, Accessed, and Entry Modified values

4: Cryptographic Hash Verification

- The system shall generate MD5 and SHA-256 hashes for each analyzed file
- The system shall compare generated hashes against known good hash databases (if provided)
- The system shall detect duplicate files across the dataset using hash comparison

5: Evidence Logging and Chain-of-Custody

- The system shall record all analysis activities with precise timestamps
- The system shall maintain chain-of-custody documentation including user identification and analysis parameters
- The system shall implement write-blocking mechanisms preventing evidence modification during analysis
- The system shall generate tamper-evident logs using cryptographic hashing with SHA-256

6: Reporting and Visualization

- The system shall generate PDF reports summarizing findings with executive summaries and detailed technical sections
- The system shall display entropy distribution graphs highlighting anomalous files
- The system shall highlight flagged files with risk ratings based on multiple detection factors
- The system shall support report customization including filtering by detection type and severity level

5. User Stories:

The following user stories capture requirements from the perspective of target users:

1: As a forensic investigator, the requirement exists to quickly scan a suspect drive for files with mismatched extensions, enabling identification of potentially hidden evidence without manual examination of thousands of files.

2: As a forensic analyst, the requirement exists to view entropy values for all files in a dataset, enabling prioritization of examination efforts toward potentially encrypted or steganographic content requiring deeper analysis.

3: As a law enforcement officer, the requirement exists for automatically generated reports with findings, enabling inclusion in case documentation and presentation to legal teams without manual compilation of technical details.

4: As a forensic examiner, the requirement exists for timeline visualizations of file modifications, enabling detection of timestamp manipulation attempts through visual identification of anomalous patterns.

5: As a corporate security analyst, the requirement exists to rapidly triage a compromised system, enabling identification of data exfiltration attempts and anti-forensic tool usage before detailed forensic imaging.

6: As a digital forensics' trainer, the requirement exists for a freely available educational tool, enabling instruction on anti-forensic detection techniques without expensive commercial software licensing.

6. Conclusion:

At the current stage, the project has successfully defined its aims, objectives, functional requirements, and methodological framework. The selection of Python and associated forensic libraries has been justified based on suitability, extensibility, and compliance with forensic best practices. The adoption of the NIST digital forensic methodology ensures that the tool's development aligns with accepted standards for evidence handling, integrity, and reporting. Additionally, a structured risk register, and agile project management approach have been established to mitigate technical, operational, and timeline-related challenges.

Although implementation and testing are ongoing, the proposed design demonstrates strong potential to improve forensic efficiency through the integration of file signature analysis, entropy-based detection, metadata examination, and cryptographic hash verification. The inclusion of visualization

and reporting capabilities further supports investigator decision-making and enhances the interpretability of analysis results during triage.

In conclusion, this report lays a solid conceptual and technical foundation for the development of an automated anti-forensic detection tool. Upon completion, the system is expected to contribute meaningfully to digital forensic practice by improving detection accuracy, reducing investigation time, and supporting forensic integrity. Future stages of the project will focus on implementation, validation using standardized forensic datasets, and evaluation of usability and detection performance against defined objectives.

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